Practice workbook · Work out the probability of a child's genotype or phenotype from a cross

Single outcomes read from a Punnett square and combined outcomes such as an affected boy or two affected children, for autosomal, sex-linked and blood-group crosses.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PROBABILITY-FROM-A-CROSS

1. Question 7 · part a.ii · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.ii only.

7 (a) Two children, M and N, are born to two sets of parents.

Child M has blood group O and child N has blood group A.

The blood groups of the two sets of parents are given in the table.

	Blood group	
	mother	**father**
1st set of parents	AB	O
2nd set of parents	B	B

(i) Explain which child belongs to which set of parents.

(4)

(ii) Calculate the probability that the next child produced by the 2nd set of parents will have blood group B.

Include a genetic diagram in your answer.

(3)

probability = ...

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1. Question 7 · part a.ii · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.ii only.

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(b) Describe how to prepare a sample of blood to be observed using a light microscope.

(5)

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(Total for Question 7 = 12 marks)

1. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

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2. Question 9 · part c.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part c.ii only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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2. Question 9 · part c.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part c.ii only.

(c) Haemophilia is a sex-linked blood disorder that reduces the ability of the blood to clot.

These are the genotypes of four offspring, P, Q, R and S.

P	Q	R	S
X^HX^h	X^hY	X^HX^H	X^HY

(i) Draw a genetic diagram to show how these offspring are produced from one set of parents.

(2)

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(ii) These parents are expecting another baby.

Determine the probability that this baby will have haemophilia.

(1)

probability = ...

(Total for Question 9 = 9 marks)

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2. Question 9 · part c.ii · 1 mark · Working and answer

Working and answer

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3. Question 6 · part b.iii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

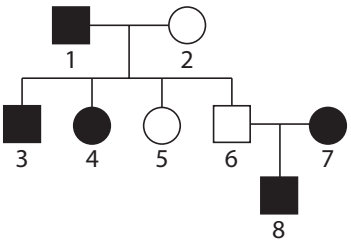
(3)

(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.



Key
- non-haemophiliac female
- non-haemophiliac male
- haemophiliac female
- haemophiliac male

X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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P 6 2 0 5 4 A 0 1 6 2 4

3. Question 6 · part b.iii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

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(ii) What is the genotype of person 5?

(1)

☒ **A** $X^h X^h$

☒ **B** $X^H X^h$

☒ **C** $X^H Y$

☒ **D** $X^h Y$

(iii) Parents 6 and 7 have another child.

Determine the probability that the child will be male with haemophilia.

(2)

probability = ...

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(iv) Explain why people with genotypes shown by persons 4 and 7 are less likely to be present in a population.

(3)

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(Total for Question 6 = 12 marks)

3. Question 6 · part b.iii · 2 marks · Working and answer

Working and answer

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4. Question 4 · part a.iv · 1 mark

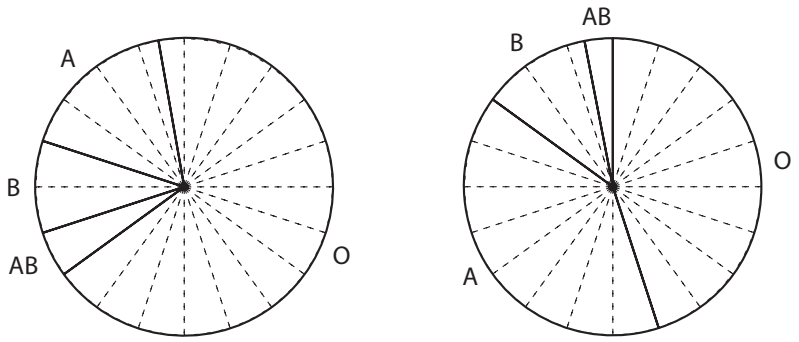
June 2022 Paper 1

Further practice · Attempt part a.iv only.

4 (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.



(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

(2)

Africa

Blood group	Percentage of blood group
O	68
A	
B	
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	
B	12
AB	

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P 7 0 9 7 3 A 0 1 2 2 4

4. Question 4 · part a.iv · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.iv only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

(3)

(iii) A male with the genotype I^AI^O and a female with the genotype I^BI^O have a child with the genotype I^OI^O.

Draw a genetic diagram to show how the child inherits the genotype I^OI^O.

(3)

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

(1)

probability = ...

4. Question 4 · part a.iv · 1 mark

June 2022 Paper 1

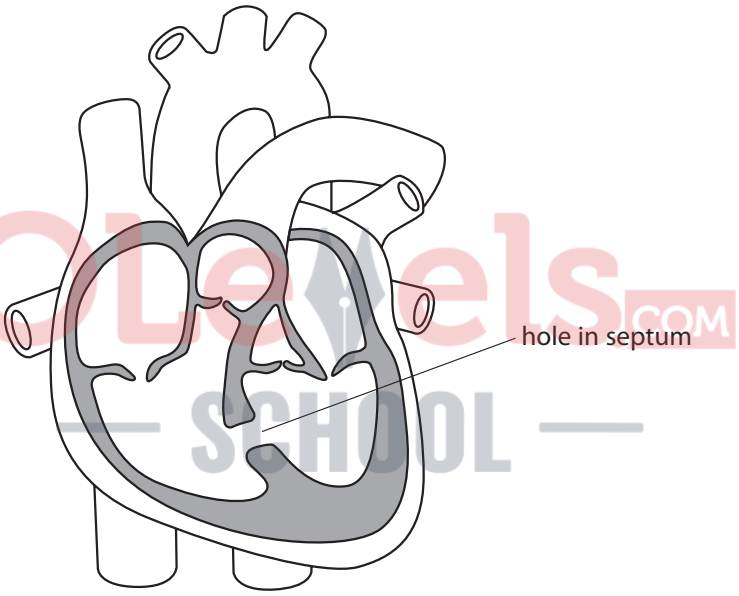
Further practice · Attempt part a.iv only.

(b) The heart pumps the blood around the body.

The diagram shows a human heart.

This heart has a hole in the septum.

This means that blood from the left side of the heart mixes with blood from the right side.



Explain why a person with a hole in their heart breathes at a greater rate than a person with a healthy heart.

(4)

(Total for Question 4 = 13 marks)

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4. Question 4 · part a.iv · 1 mark · Working and answer

Working and answer

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5. Question 6 · part b.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

6 Blood will clot when there is damage to the skin, such as a cut.

(a) (i) State two reasons why it is important for the blood to clot.

(2)

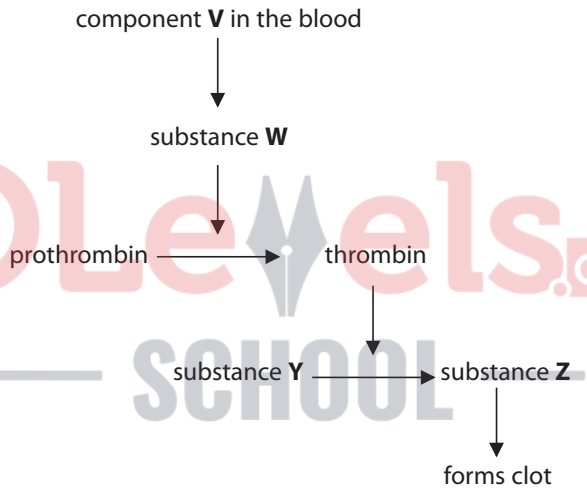
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2 .....

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(ii) The diagram shows some of the stages in the process of blood clotting.



Use words from the box to identify V, W, Y and Z.

thrombokinase enzyme		white blood cells		platelets
haemoglobin	fibrinogen	fibrin	antigens	antibody

(4)

V .....

W .....

Y .....

Z .....

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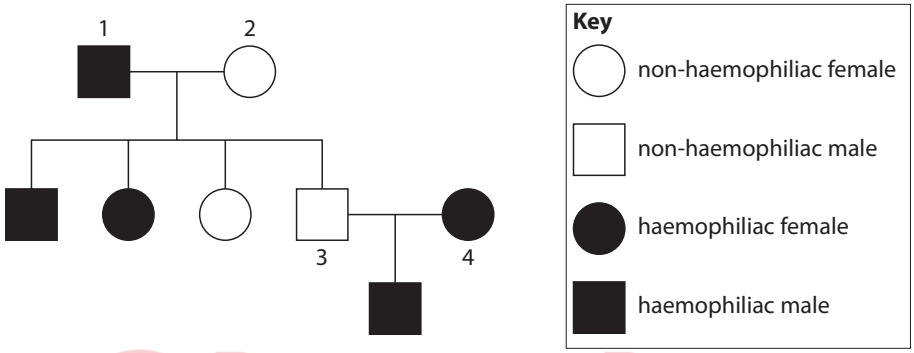
5. Question 6 · part b.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

(b) Haemophilia is an inherited condition where the blood of a person does not clot.

The diagram shows a family pedigree. Some members of the family have the condition.



(i) Deduce the genotypes of family members 1 and 2 showing how you obtained your answer.

(5)

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5. Question 6 · part b.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Male 3 and female 4 have a second child.

Deduce the probability that this child will be a boy with haemophilia showing how you obtained your answer.

(4)

(Total for Question 6 = 15 marks)

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5. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

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6. Question 8 · part b.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

8 (a) Alkaptonuria is a condition caused by a gene mutation that produces a recessive allele.

People who have this condition are unable to fully break down two amino acids found in proteins. Instead, an intermediate compound is formed which gives a dark colour to the urine.

This intermediate compound causes damage to cartilage and heart valves.

(i) Explain how amino acids are normally metabolised.

(3)

(ii) A person with alkaptonuria produces a dark coloured urine.

Describe what other symptoms they may develop.

(3)

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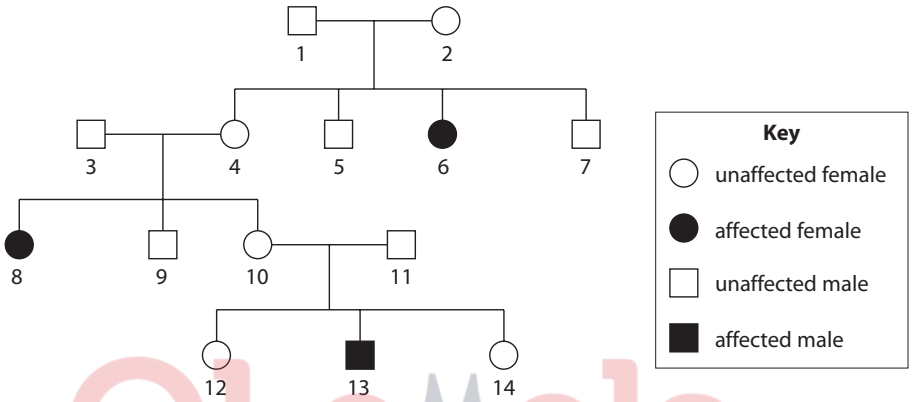
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6. Question 8 · part b.ii · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

(b) The diagram shows the pedigree of a family affected by alkaptonuria.



(i) Use the family pedigree to explain whether the alkaptonuria condition is sex-linked.

(4)

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6. Question 8 · part b.ii · 3 marks

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Further practice · Attempt part b.ii only.

(ii) Parents 10 and 11 have another child.

Calculate the probability that this child would be an affected male.

Explain your answer.

(3)

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(Total for Question 8 = 13 marks)

6. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part b.ii only.

8 Haemophilia is a sex-linked condition caused by a recessive allele X^h instead of the usual X^H.

Haemophilia is the result of a mutation.

(a) Explain what is meant by the term **mutation**.

(3)

(b) (i) A man without haemophilia and a woman who is a carrier of haemophilia have a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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7. Question 8 · part b.ii · 3 marks

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Further practice · Attempt part b.ii only.

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(ii) The man and woman have a second child.

Calculate the probability that both children will have haemophilia.

(3)

(c) Explain why the man cannot be a carrier of haemophilia.

(5)

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(Total for Question 8 = 15 marks)

7. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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8. Question 8 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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P 6 6 4 3 3 A 0 2 0 2 4

8. Question 8 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Calculate the probability that these parents will produce a boy with the condition.

(3)

probability = ...

(c) White forelock is a result of a mutation in a molecule of DNA.

Describe how this mutation causes the white forelock phenotype.

(4)

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(Total for Question 8 = 14 marks)

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8. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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9. Question 7 · part b.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

7 Huntington's chorea is a dominant genetic condition that results in the nervous system not working correctly. It is not a sex-linked condition.

Diagram 1 shows a family pedigree. Some members of the family have the condition.

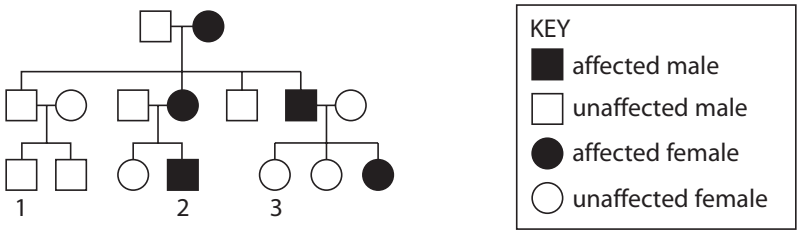


Diagram 1

(a) (i) Evaluate the evidence in the pedigree that supports the statement that this is not a sex-linked condition.

(2)

(ii) Evaluate the evidence in the pedigree to support the statement that this is a dominant condition.

(2)

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9. Question 7 · part b.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

(b) (i) A man has the same genotype as person 1 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

These two people are expecting a child.

Calculate the probability that the child would be a boy not showing the Huntington's condition, giving reasons for your answer.

(3)

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9. Question 7 · part b.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

(ii) A man has the same genotype as person 2 in the pedigree in Diagram 1.

A woman has the same genotype as person 3 in the pedigree in Diagram 1.

Explain the possible phenotypes and genotypes, with respect to the Huntington's condition, of any offspring they produce.

(6)

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(Total for Question 7 = 13 marks)

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9. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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10. Question 5 · part c.ii · 4 marks

November 2023 Paper 2

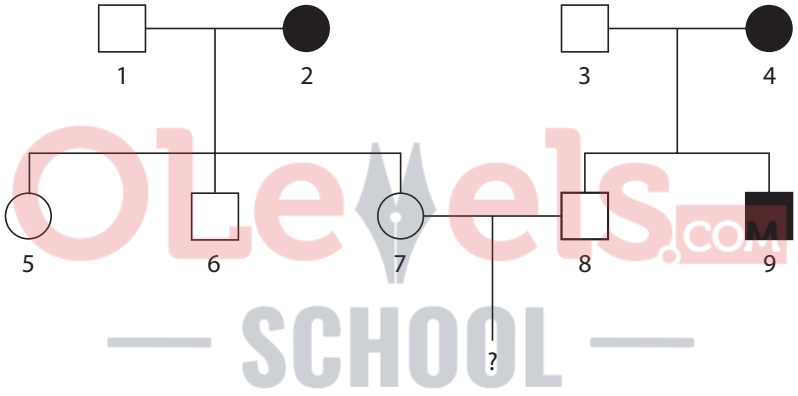
Further practice · Attempt part c.ii only.

5 Sickle cell anaemia is a condition caused by a recessive allele.

(a) Explain what is meant by the term **recessive allele**.

(2)

(b) The diagram shows a family pedigree. Some family members have sickle cell anaemia.



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Key

- Male with sickle cell anaemia
- Male without sickle cell anaemia
- Female with sickle cell anaemia
- Female without sickle cell anaemia

(i) Give two people who are homozygous recessive.

(2)

(ii) Give the genotype of person 6.

Use the symbol **A** for the non-sickle cell allele and the symbol **a** for the sickle cell allele.

(1)

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10. Question 5 · part c.ii · 4 marks

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Further practice · Attempt part c.ii only.

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(c) (i) People 7 and 8 in the pedigree produce a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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(ii) Calculate the probability that this child will be a boy with sickle cell anaemia.

(4)

probability = ...

(Total for Question 5 = 13 marks)

10. Question 5 · part c.ii · 4 marks · Working and answer

Working and answer

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11. Question 5 · part b.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

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5 (a) (i) Describe what is meant by the term **mutation**.

(2)

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

(3)

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11. Question 5 · part b.iii · 3 marks

November 2024 Paper 2

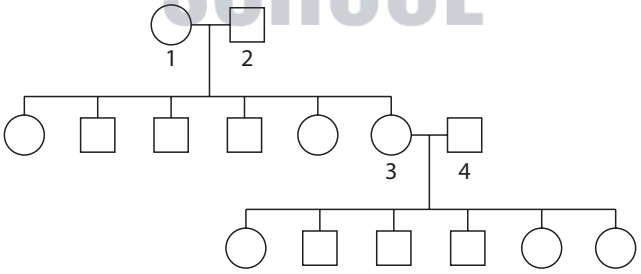
Further practice · Attempt part b.iii only.

(b) (i) The condition known as hairy ears is thought to be caused by a mutation of an allele on the Y chromosome.

Explain how the genetics of this sex-linked condition differ from the genetics of haemophilia.

(2)

(ii) The diagram is a family pedigree showing the inheritance of the hairy ears mutation.



Person 1 is a female without the condition and person 2 is a male with the condition.

State the genotype of person 1 and the genotype of person 2.

Use H to represent the allele for hairy ears.

(2)

person 1

person 2

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11. Question 5 · part b.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part b.iii only.

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(iii) Neither person 3 nor person 4 has the condition.

Explain why the probability of their next child being a girl without hairy ears is 50%.

(3)

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(Total for Question 5 = 12 marks)

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Working and answer